Time & Space Complexity

→ Linear Search :-

```
bool isPresent (int *arr, int n, int value)
{
    for (int i=0; i<n; i++)
    {
        if (arr[i] == value)
            return true;
    }
    return false;
}
```

T.C → O(n) (n → arr size)

S.C → O(1)

→ Recursion:- Fac / Power / Counting

Factorial → T.C → time required as function of input

$f(n)$

```
int fact (int n)
{
    // Base Calc
    if (n==0)
        return 1;

    // Rec. Call
    return n * fact(n-1);
}
```

K_1

$n * fact(n-1)$ → K_1 , $T(n-1)$

→ T.C → ?

① Try -> Eq -> Recurrence

$$f(n) = n * f(n-1)$$

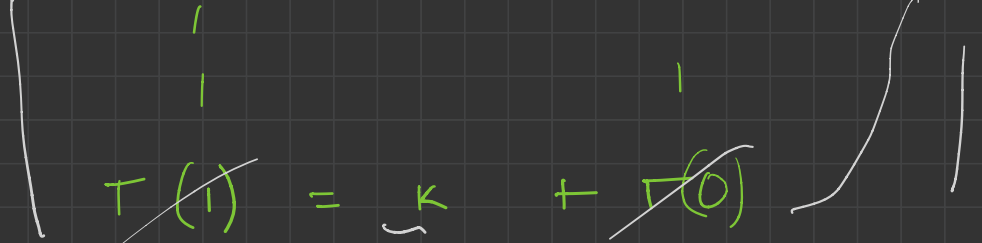
$$T(n) = K_1 + K_2 + T(n-1)$$

$$T(n) = K + T(n-1)$$

$$T(n-1) = K + T(n-2)$$

$$T(n-2) = K + T(n-3)$$

n times



$T(1) = K + T(0)$

$T(0) = K_1$

This denotes base case. It is T(0) because The smallest input size for which the factorial calculation terminates is n=0, because 0! is defined as 1.

$T(n) = n * K + K_1$

$T(n) = Kn + K_1$

$= \alpha$

$T(n) = Kn$

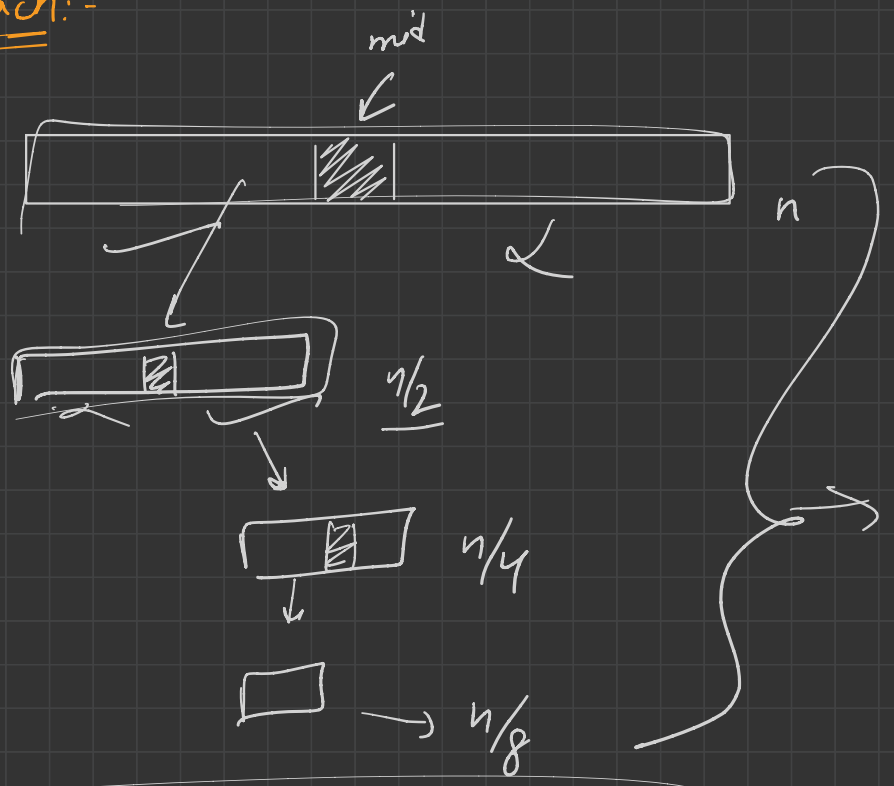
$T(n) = n$

$O(n)$

Factorial

T.C → $O(n)$

Binary Search:-



$$f(n) = \underbrace{xyz}_{} + f(n/2)$$

$$T(n) = K_1 + K_2 + T(n/2)$$

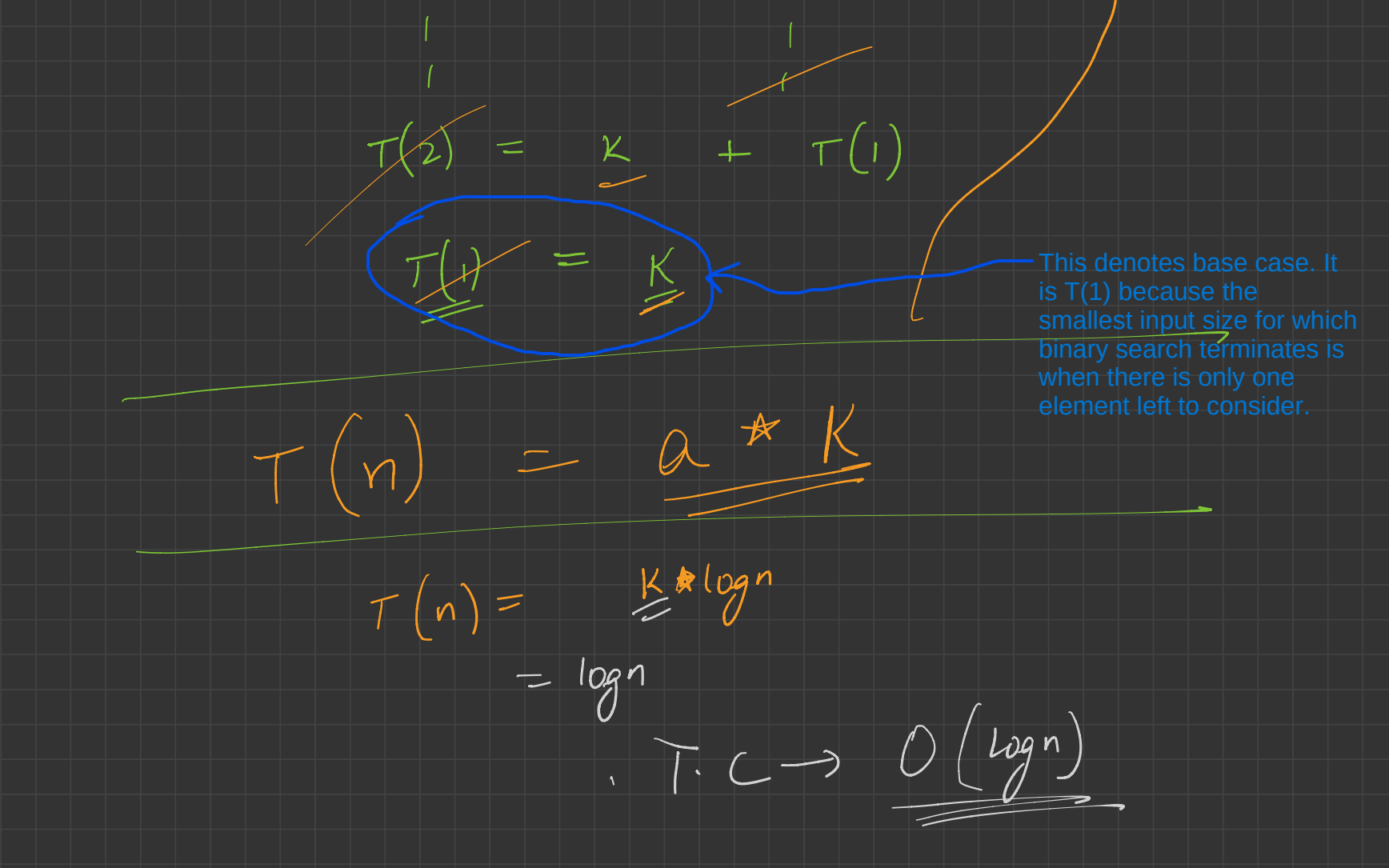
$$K$$

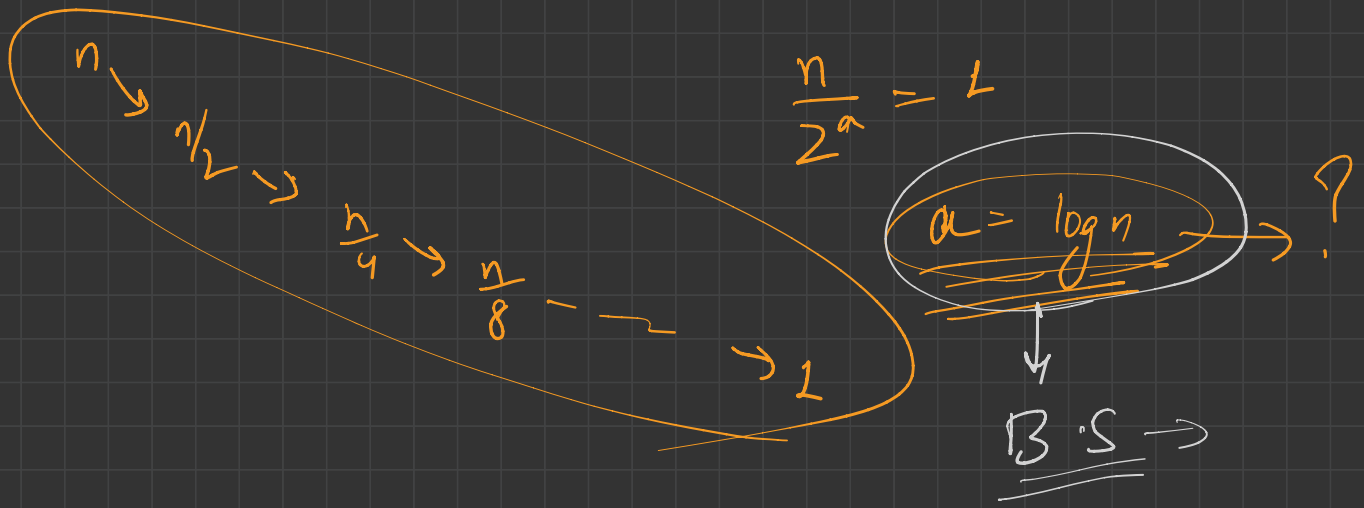
$$T(n) = K + T(n/2)$$

$$T\left(\frac{n}{2}\right) = K + T\left(\frac{n}{4}\right)$$

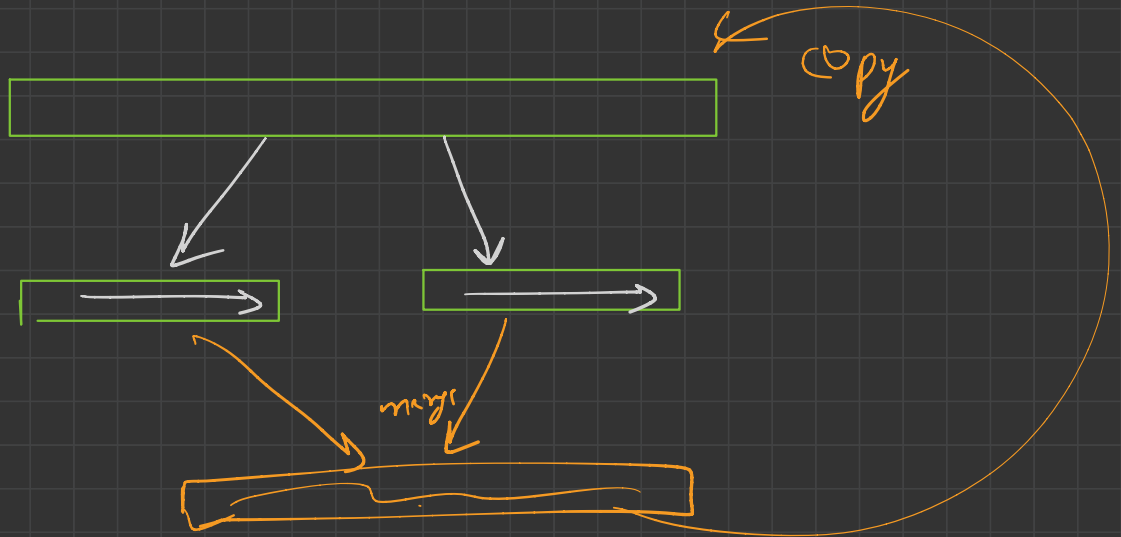
$$T\left(\frac{n}{4}\right) = K + T\left(\frac{n}{8}\right)$$

a times


T(2) = K + T(1)
T(1) = K
This denotes base case. It is T(1) because the smallest input size for which binary search terminates is when there is only one element left to consider.
T(n) = a * K
T(n) = K * log n
= log n
∴ T.C → O(log n)


n → n/2 → n/4 → n/8 → 1
n/2^a = 1
a = log n
?
B.S →

→ Merge Sort



Steps

→ break 2 array left & right

→ Rec → sort (left / Right)

→ new array → merge → K3 n

→ copy new array content → original array → K4 n

left

right part

$$T(n) = K_1 + K_2 + T\left(\frac{n}{2}\right) + T\left(\frac{n}{2}\right) + K_3 n + K_4 n$$

$K_1 + K_2 = K$

$$T(n) = K + 2T\left(\frac{n}{2}\right) + n(K_3 + K_4)$$

$K_3 + K_4 = K_5$

$$T(n) = K + 2T\left(\frac{n}{2}\right) + nK_5$$

Here, K is ignored as it is nothing compared to n.

$K \approx 0$

$$T(n) = 2T\left(\frac{n}{2}\right) + nK_5$$

$$T(n) = 2T\left(\frac{n}{2}\right) + nk$$

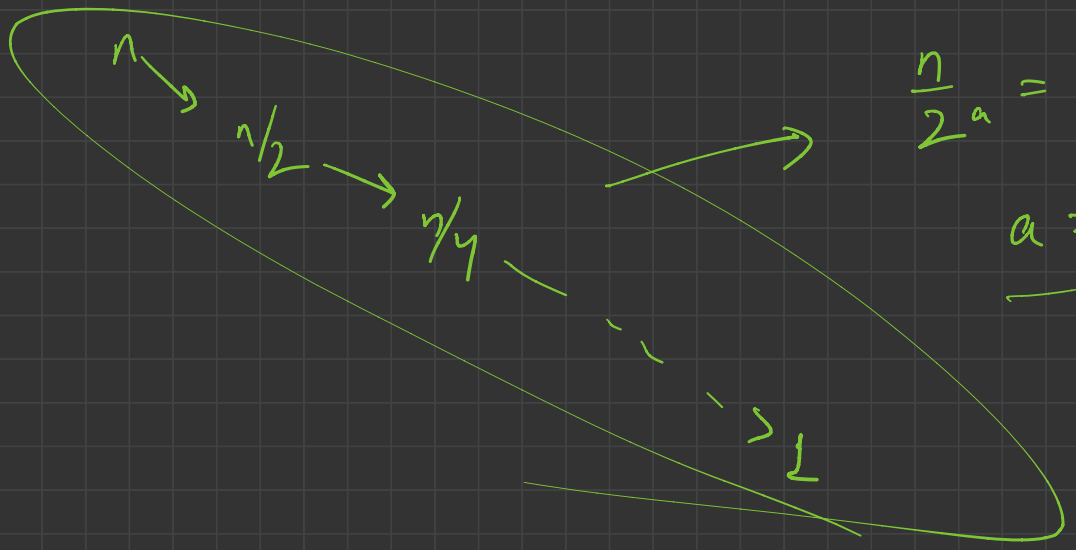
$$2\times\left[T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + \frac{n}{2}k\right] = 4T\left(\frac{n}{4}\right) + nk$$

$$4\times\left[T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right) + \frac{n}{4}k\right] = 8T\left(\frac{n}{8}\right) + nk$$

a fimes

$$T'(1) = K$$

$$T(n) = a * nK$$



$$\frac{n}{2^a} = 1$$

$$a = \log n$$

$$T(n) = an$$

$$= \log n \times n$$

$$= n \log n \rightarrow O(n \log n)$$

T.C

→ Fibonacci Series

```
fib( int n)
{
    // B.C
    if (n == 0 || n == 1)
        return n;

    // R.C
    return fib(n-1) + fib(n-2);
}
```

K_1 → T.C → $O(2^n)$

K_2

$$T(n) = K_1 + K_2 + T(n-1) + T(n-2)$$
$$= K + T(n-1) + T(n-2)$$

$$T(n) = K + T(n-1) + T(n-2)$$

$$T(n-1) = K + T(n-2) + T(n-3)$$

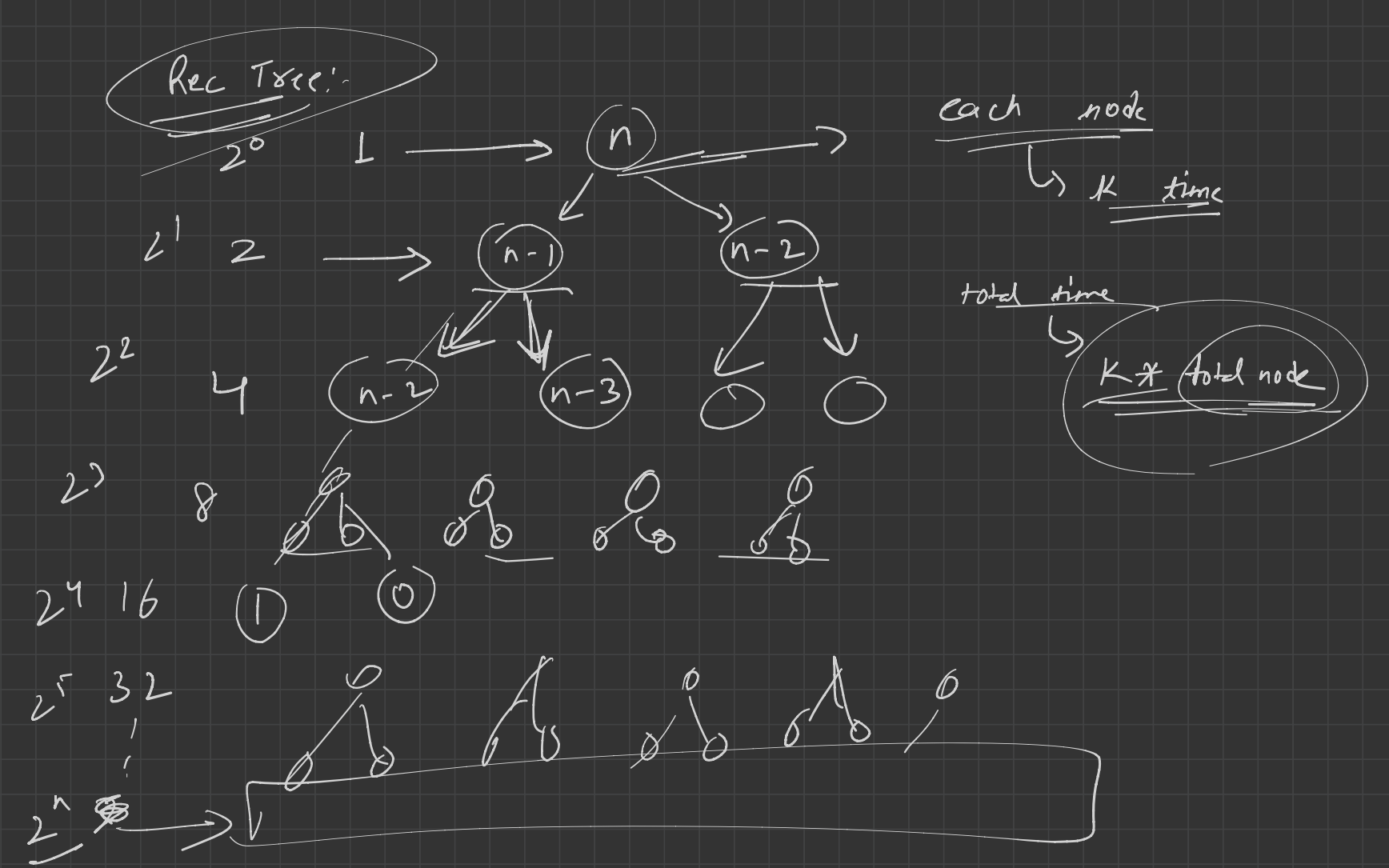
$$T(n-2) = K + T(n-3) + T(n-4)$$

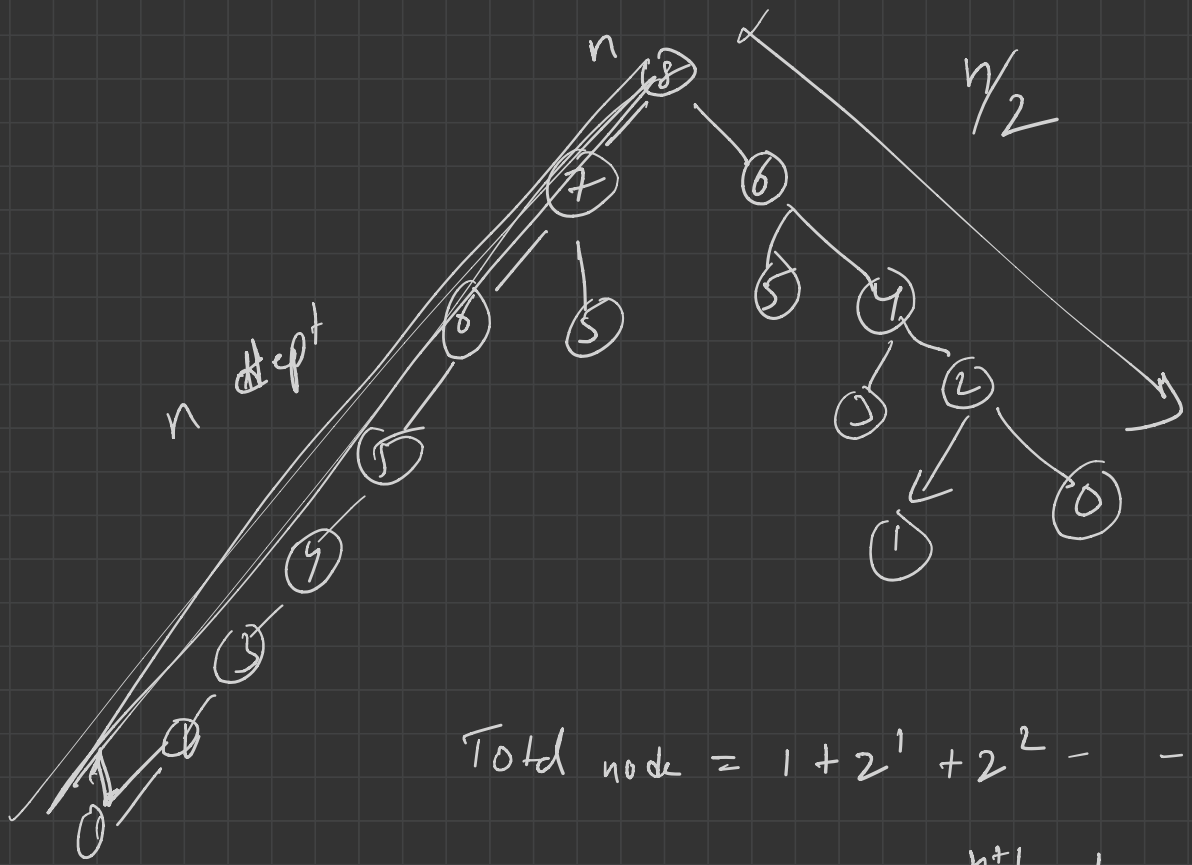
Solve - ?

by Hw

$$T(1) = K_1$$

$$T(0) = K_2$$


Rec Tree:-
2^0 1
n
each node
K time
2^1 2
n-1
n-2
total time
K * total node
2^2 4
n-2
n-3
2^3 8
2^4 16
1
0
2^5 32
2^n



Total node $= 1 + 2^1 + 2^2 - - - 2^n$

$= 2^{n+1} - 1$

$$T.C = K * (2^{n+1} - 1)$$

$$= K \times 2^{n+1}$$

$$= K \times 2 \times 2^{n}$$

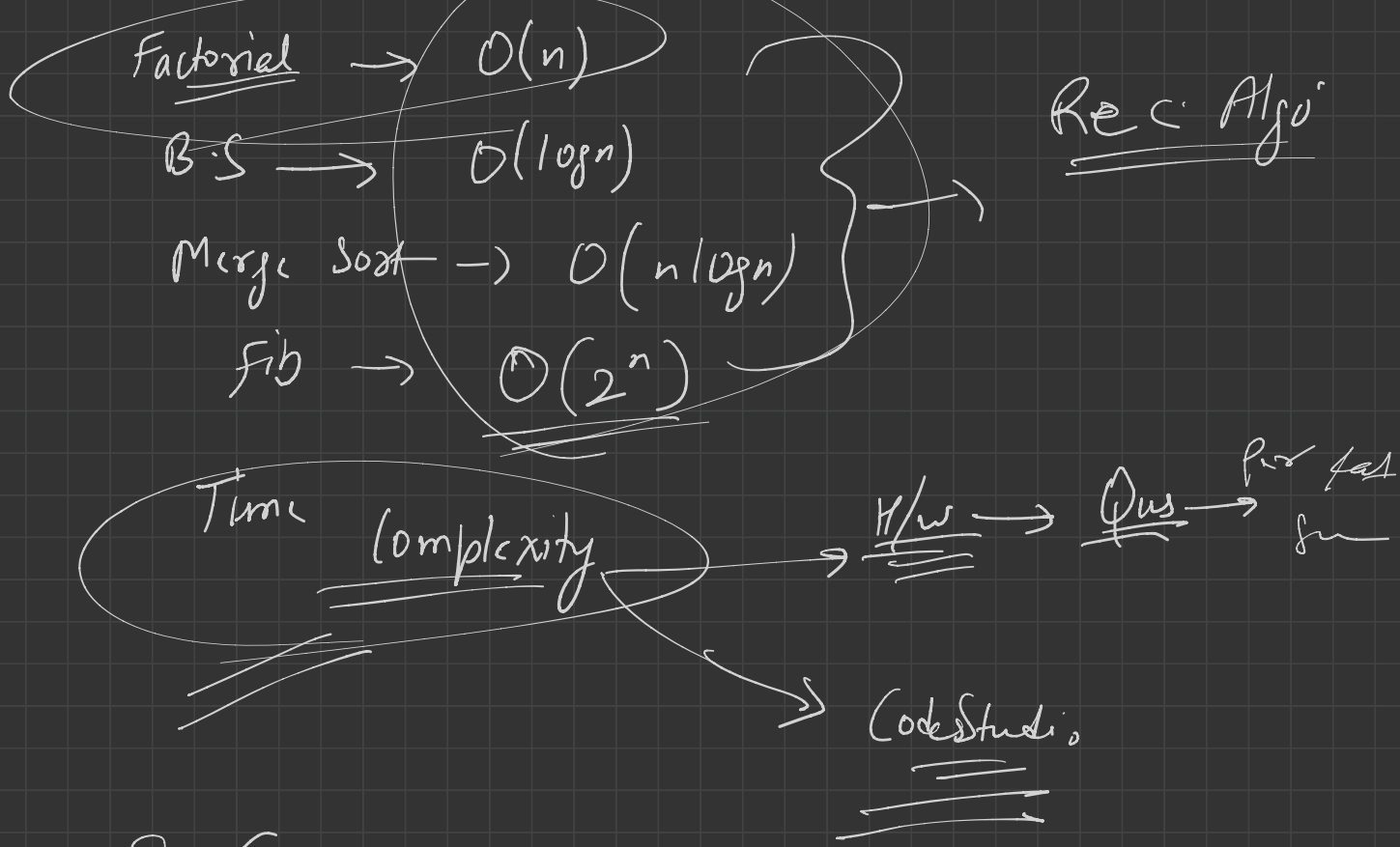
$$= K \times 2^{n}$$

$$= 2^{n}$$

Fib

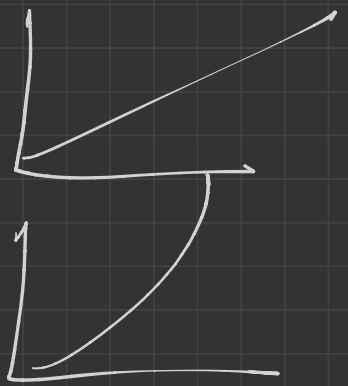
T.C $\longrightarrow O(2^n)$

exponential

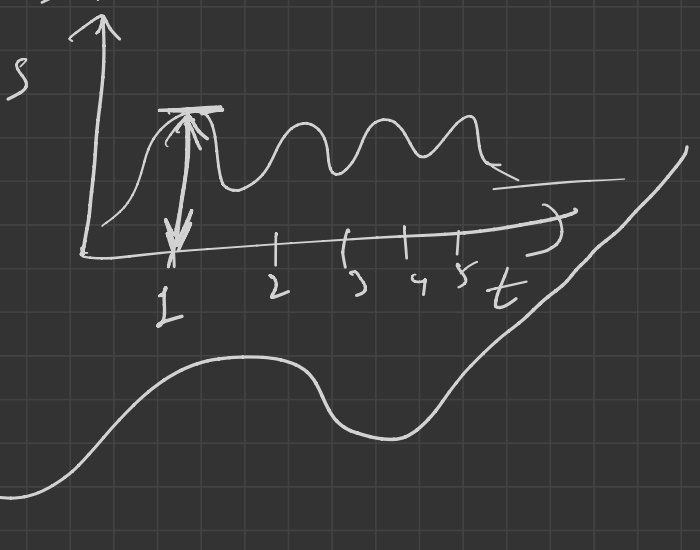

Factorial → O(n)
B.S → O(logn)
Merge Sort → O(nlogn)
Fib → O(2^n)
Rec Algo
Time complexity
H/w → Qus → For fast
CodeStudio
S.C

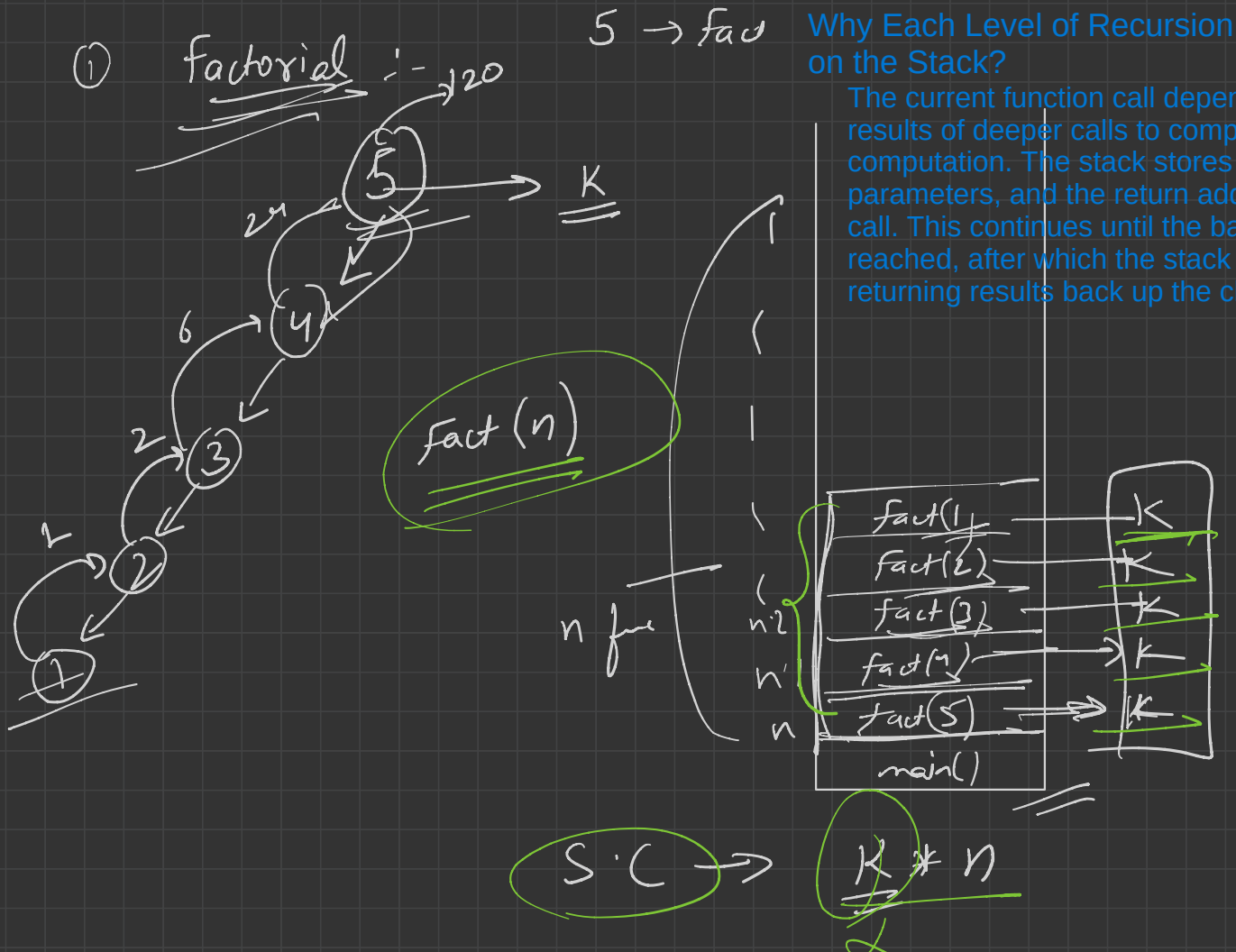
Space Complexity :- space required as $f(n)$

T.C



max. space req. at any instant

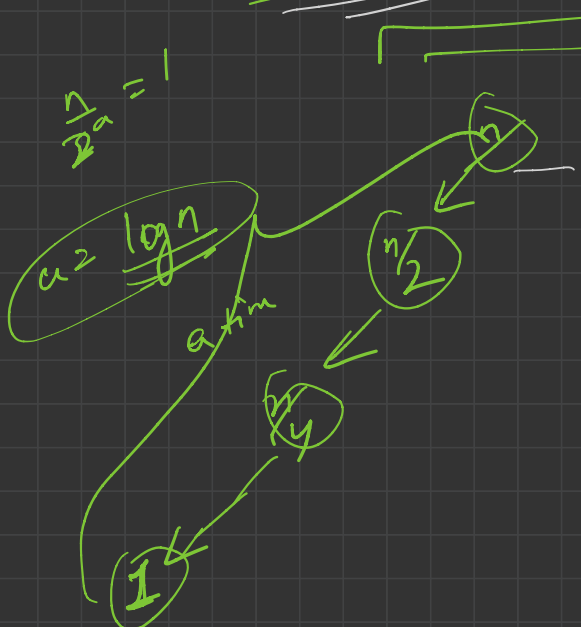


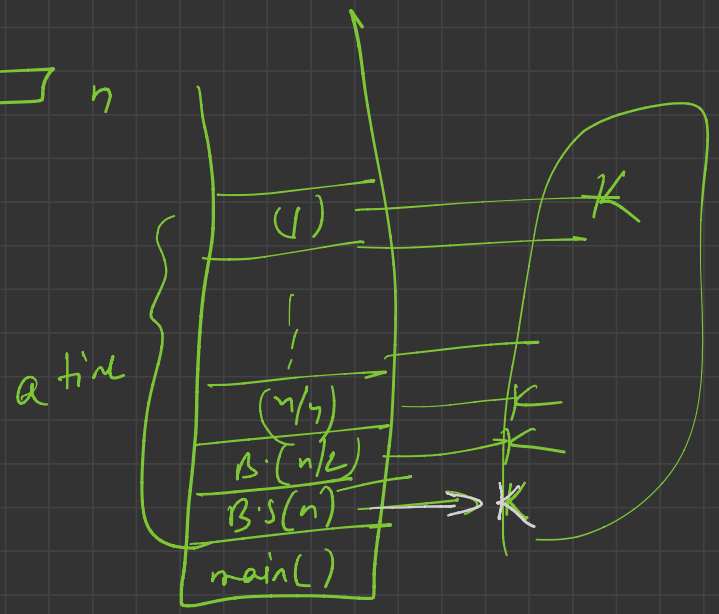

① Factorial :-
5 → fact
Why Each Level of Recursion Accumulates on the Stack?
The current function call depends on the results of deeper calls to complete its computation. The stack stores local variables, parameters, and the return address for each call. This continues until the base case is reached, after which the stack unwinds, returning results back up the chain.
120
5
K
24
6
4
2
3
1
2
1
Fact (n)
1
(
1
\
(
n-2
n-1
n
n fact
fact(1)
fact(2)
fact(3)
fact(4)
fact(5)
main()
K
K
K
K
K
S.C → K * n

factorial

↳ O(n)

→ Binary Search:-

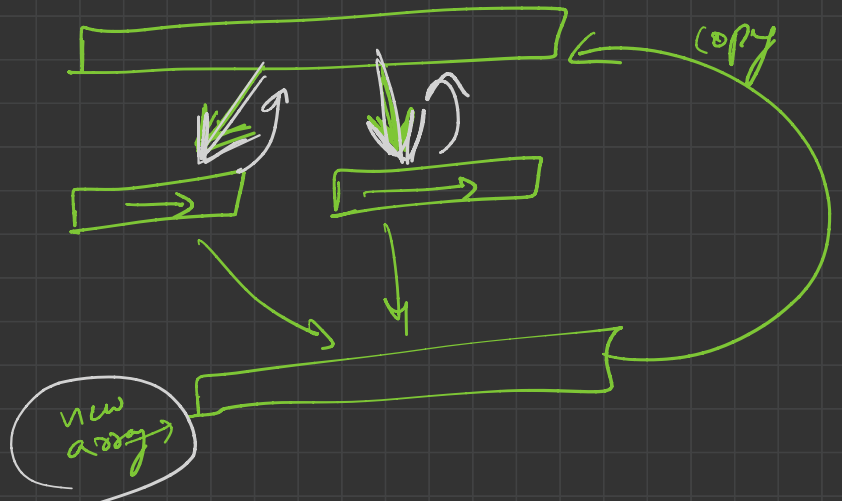


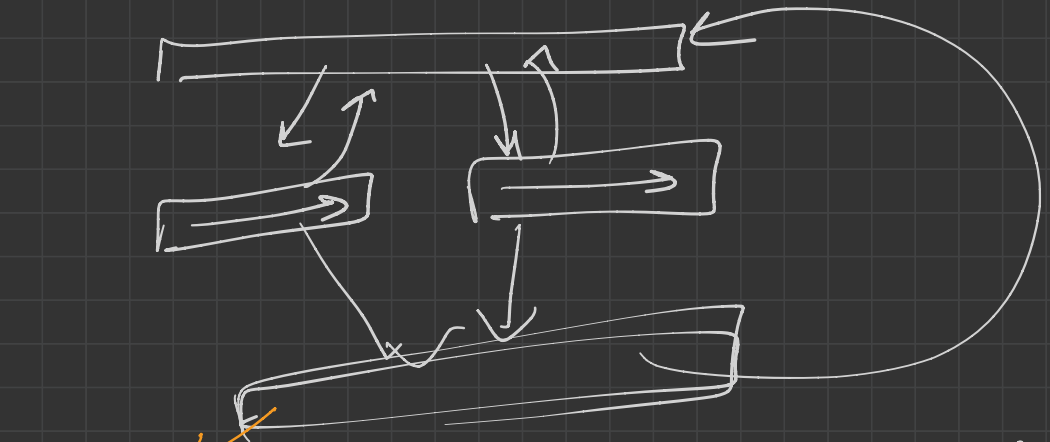


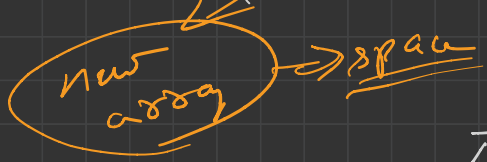
S.C → $\frac{K}{2} * \log n$

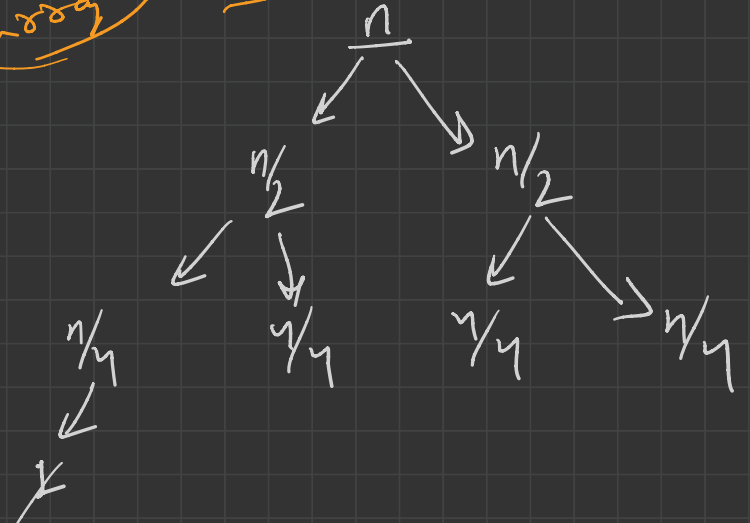
$= O(\log n)$

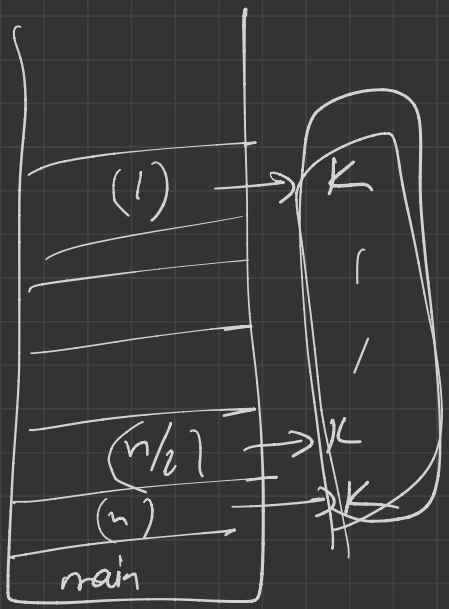
③ Merge sort →

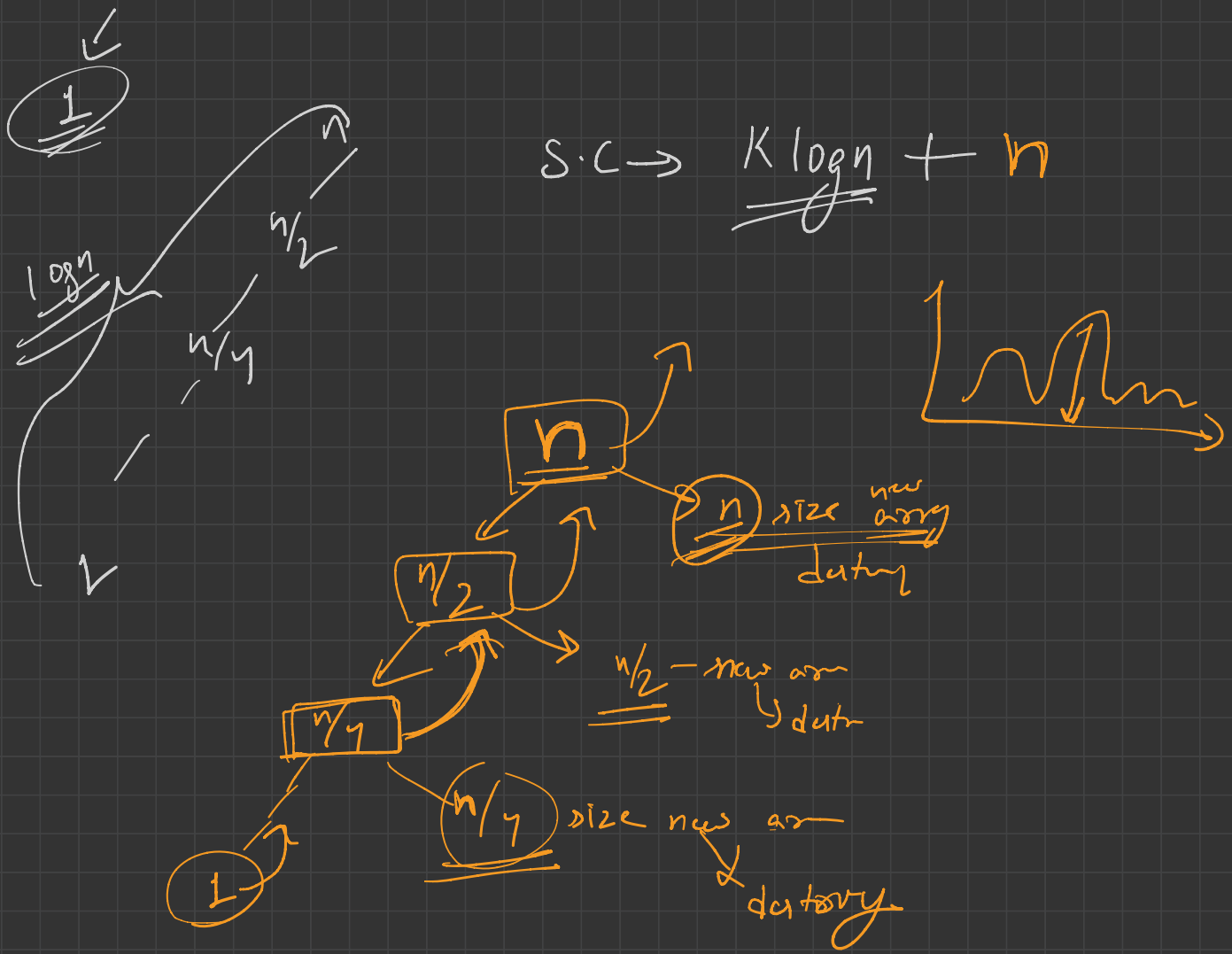





New array
space


n
n/2
n/2
n/4
n/4
n/4
n/4
K


(1)
K
(n/2)
K
(n)
K
main


1
n
n/2
n/4
1
log n
S.C → K log n + n
n
n
size new array
destroy
n/2
n/2 — new arr
destroy
n/4
n/4
size new arr
destroy
1

S.C → $K \log n + n$

$n >>> \log n$

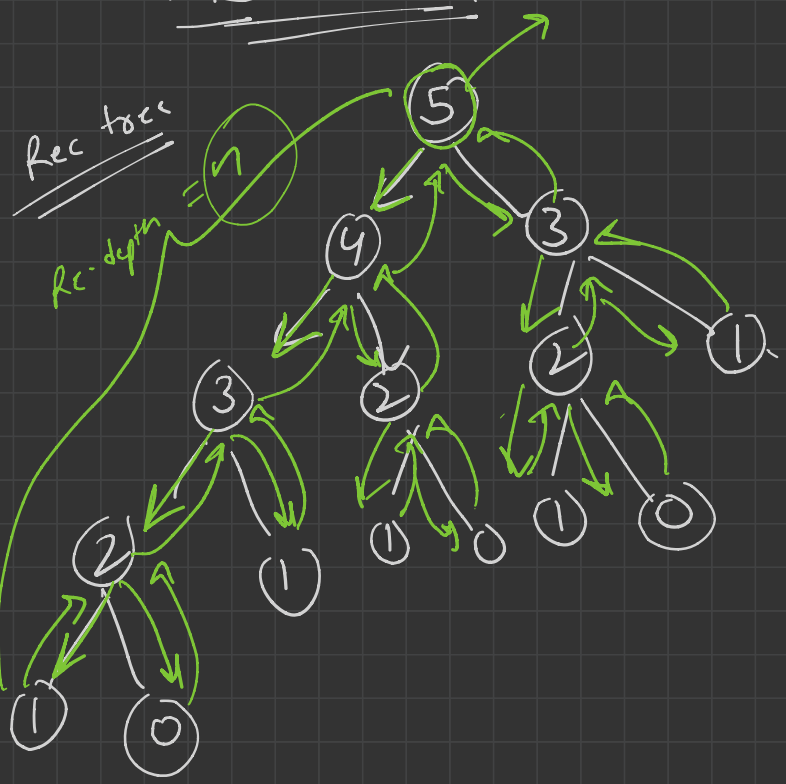
S.C → $O(n)$

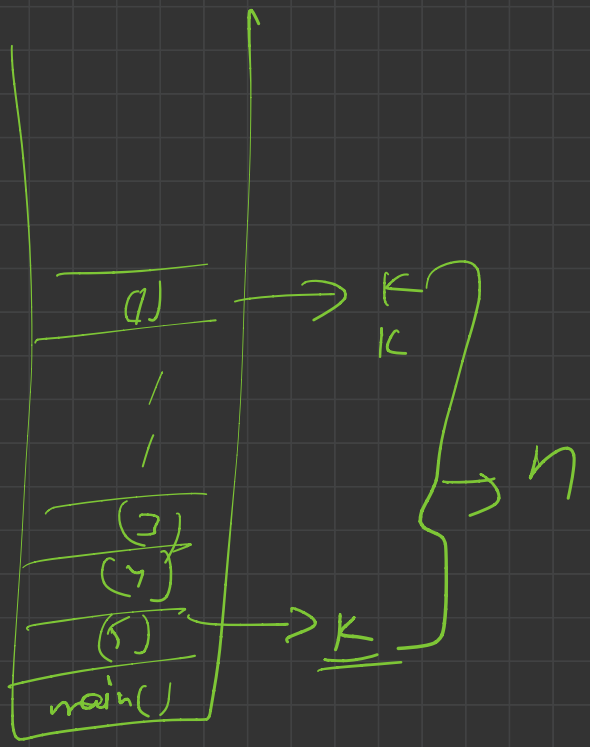
Recc Merge sort

$n = 1024$

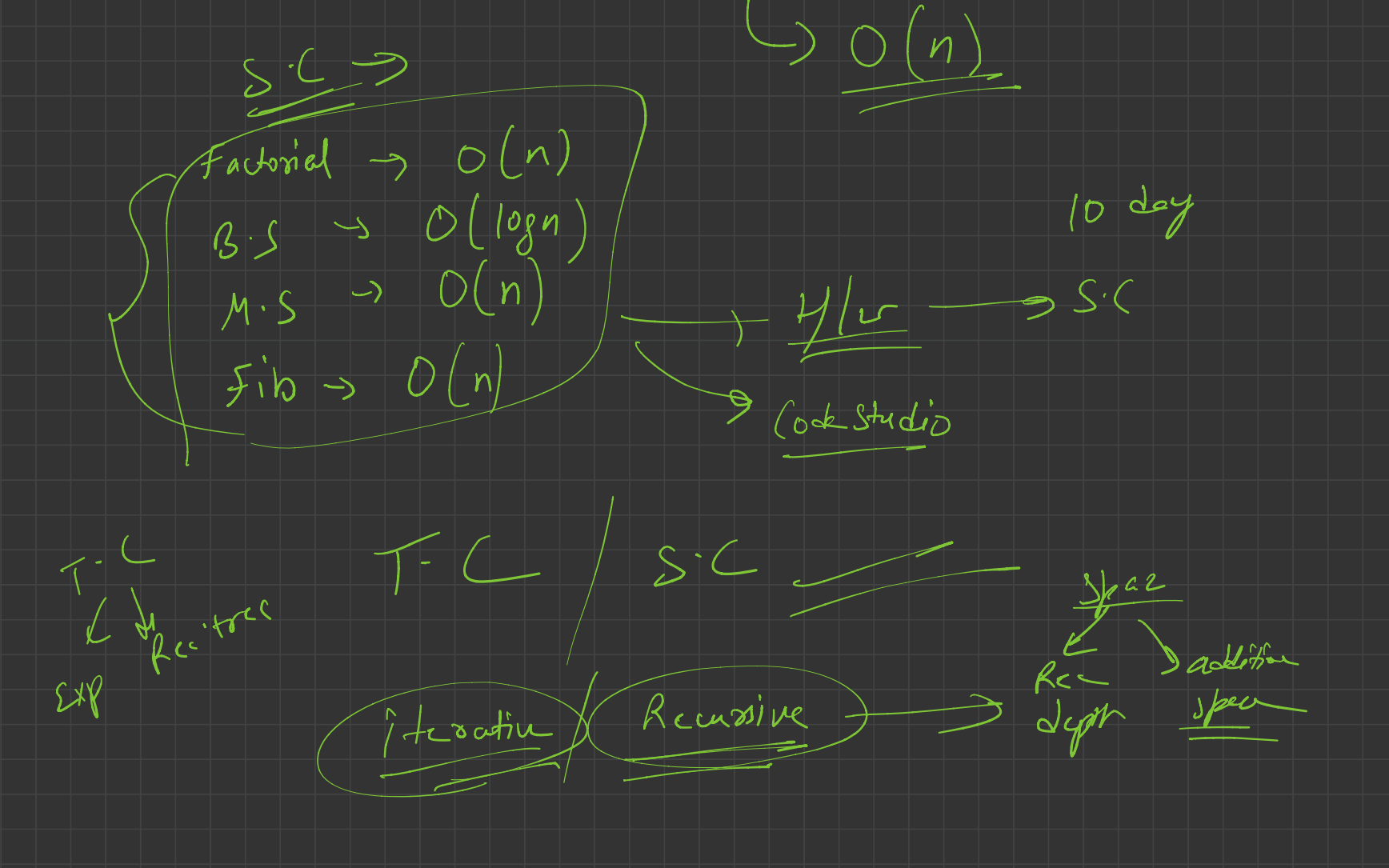
$\log n = 10$

Fib Series:-





S.C → n*k


O(n)
S.C →
Factorial → O(n)
B.S → O(log n)
M.S → O(n)
Fib → O(n)
10 day
H/Ir
S.C
Codestudio
T.C / S.C
T.C
Recursive
EXP
Iterative
Recursive
Space
Rec
depth
addition
space